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Readability of patient education materials in hand and wrist surgery: Can ChatGPT make a difference?

Ailbhe Quigley [a,*], Tiarán Ó Doinn [b], Louise Kent [a], James M. Broderick [b]

[a] Department of Trauma & Orthopaedics, Connolly Hospital Blanchardstown, Ireland
[b] Department of Trauma & Orthopaedics, St Vincent's University Hospital, Ireland

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ABSTRACT

Background: The internet is a common resource among orthopaedic patients. Many orthopaedic Patient Education Materials (PEMs) are written above the recommended 6th-grade reading level. This study aims to evaluate the readability of online hand and wrist PEMs and determine whether AI can enhance their readability.

Methods: 135 hand and wrist surgery PEMs were identified from the British Society for Surgery of the Hand, the American Society for Surgery of the Hand, and the American Academy of Orthopaedic Surgeons websites. Readability was assessed using eight formulae. The mean Reading Grade Level (RGL) was compared to the 6th and 8th-grade reading levels. 27 articles were input into ChatGPT with the following command: "Rewrite the following patient education materials using a 6th-grade reading level. The material should be ready to publish in the patient education section of a healthcare website." The readability was then reassessed.

Results: The mean RGL of the 135 PEMs was 9.89 (range = 7.2–15), significantly higher than the 6th-grade level by 3.89 grade levels (95 % CI, 3.7–8; $P < 0.001$) and the 8th-grade level by 1.89 grade levels (95 % CI, 1.7–2.08; $P < 0.001$). The 27 pre-simplification articles had a mean RGL of 10.4. The ChatGPT-simplified PEMs' mean RGL was 8.54, exceeding the requested 6th-grade level by 2.54 grade levels (95 % CI, 2.13–2.96; $P < 0.001$). The simplified articles' mean RGL was significantly lower than the original articles (95 % CI, 1.2–2.49; $p < 0.001$).

Conclusion: Hand and wrist PEMs from leading healthcare organisations are written above the recommended levels. ChatGPT shows potential as an accessible tool to improve readability. Integrating AI into the creation and review process of PEMs could be a valuable strategy for improving readability.

1. Introduction

The internet has become a popular source of information for individuals seeking information on health conditions, thereby playing a crucial role in how patients engage with their healthcare.[1,2] As more patients turn to the internet for health information, the quality and accessibility of online Patient Education Materials (PEMs) are as important as ever with the shared decision-making model being embraced by modern healthcare. This model places emphasis on collaboration between doctors and patients, ensuring the patient is actively involved in their treatment decisions. For this process to be effective, patients must be well informed about their condition as well as treatment options and the benefits and risks of each.[3] Consequently, online PEMs have gained increasing importance in empowering patients to make informed decisions regarding their health. The majority of patients use the internet as a source for medical information.[1,4]

Online PEMs are helpful in allowing patients to access information on health conditions at their convenience, enabling them to engage with their health condition outside of the clinical setting and consider the options available to them. However, the efficacy of these materials is dependent on their readability, which is an important factor influencing how well patients understand the information provided. Readability refers to how easily a piece of text can be read and comprehended by the intended audience. In the context of health education, it should be ensured that PEMs are accessible to individuals with varying literacy levels, especially patients with low health literacy, a factor strongly correlated with patient outcomes.[5–8] Health literacy refers to "the degree in which an individual is able to acquire, process, and comprehend information in a way to be actively involved in their health decisions".[9]

Numerous formulae have been developed to assess the readability of written materials, most of which provide a Reading Grade Level (RGL) as the unit of measurement. RGL indicates the minimum education level

* Corresponding author. Department of Trauma & Orthopaedics, Connolly Hospital Blanchardstown, Mill Rd, Abbotstown, Dublin, D15 X40D, Ireland.
E-mail address: ailbhe.quigley@hse.ie (A. Quigley).

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(based on the US education system) required to understand the text. Given that the average RGL of adults in the United States is the 8th grade level,[5] health information written above this level risks excluding a significant portion of the population, particularly those with limited literacy skills. The National Institute of Health (NIH) has previously recommended a 6th-7th grade reading level for health materials.[10] Although they no longer specify this exact level on their website, they continue to advocate for the use of plain language and designing materials for those with limited health literacy. The American Medical Association (AMA) has similarly recommended that PEMs should be written at or below the 6th grade level,[11] whereas the Agency for Healthcare Research and Quality (AHRQ) recommends a 5th or 6th grade level.[5]

Despite these guidelines, a growing body of research has shown that many online PEMs, including those related to Orthopaedic conditions, are written at a reading level that exceeds the recommended levels and often exceeds the average literacy level of the general population.[12–17] This discrepancy provides a substantial barrier to patient understanding and engagement, particularly for those with lower health literacy. As it has been shown that there is a strong correlation between health literacy and poorer mental and physical health as well as poorer patient outcomes,[6,18] many studies conclude that further work must be done to improve the readability to enhance patients' health literacy.[12–14,19–21]

Firstly, this study aims to assess the current readability of PEMs in hand and wrist surgery from 3 leading Orthopaedic websites: the British Society for Surgery of the Hand (BSSH), the American Society for Surgery of the Hand (ASSH), and the American Academy of Orthopaedic Surgeons (AAOS). Secondly, we aim to examine the ability of AI to simplify PEMs, to more closely approximate the recommended 6th grade reading level.

2. Materials and methods

In June 2024, a search was conducted of the PEMs of the American Academy of Orthopaedic Surgeons (AAOS) (www.aaos.org), the American Society for Surgery of the Hand (ASSH) (www.assh.org), and the British Society for Surgery of the Hand (BSSH) (www.bssh.ac.uk). The British Orthopaedic Association (BOA) (www.boa.ac.uk) website was reviewed, however, their information on the Orthopaedic conditions section for patients was directed to the BSSH patient education section. PEMs related to conditions of the hand and wrist were identified. Articles were excluded if the content was not written in English, if it was not directed towards patients or if the material was primarily presented in video, graphic, table, or list format.

The text from each article was copied and pasted into a Microsoft Word document (Microsoft, Redmond, WA). Any figures, figure captions, advertisements, hyperlinks, copyright notices and text that was not directly related to the PEM were excluded. The re-formatted patient education resources were subsequently input into Readability Studio Professional Edition Version 2019 (Oleander Software Ltd.) for readability analysis. The readability of the articles was assessed using eight different formulae, including the Flesch-Kincaid Reading Grade Level (FKGL); Raygor Estimate; Fry; SMOG; Coleman-Liau; FORECAST, Gunning Fog and the Flesch Reading Ease Score (FRES). All RGLs were reported as a United States grade level, denoting the years of education required based on the US educational system to easily read and understand a piece of text. All formulae generate a RGL except the FRES formula which calculates the readability expressed as an index score which is based on sentence length and number of syllables. The index score ranges from 0 to 100, with higher scores denoting easier readability. The number of articles with a RGL less than or equal to the 6th grade and 8th grade level was ascertained. The overall mean RGL of all articles was compared with 6th grade and 8th grade reading levels, using one-sample t-tests. Cumulative mean RGLs for the three website's articles were compared using a one-way analysis of variance (ANOVA).

Nine articles from each of the three websites were selected to be input into ChatGPT 4o (OpenAI, San Francisco, CA; version 4o) for simplification. The topics were consistent across the three websites and included carpal tunnel syndrome; congenital hand conditions; de Quervain's syndrome; Dupuytren's disease; flexor tendon injury; ganglion cyst; mallet finger; basal thumb arthritis; and trigger finger. The mean RGL of these 27 articles was determined and was compared with 6th grade and 8th grade reading levels, using one-sample t-tests. Each of the 27 articles was input into ChatGPT 4o with the following command: "Rewrite the following patient education materials using a 6th grade reading level. The material should be ready to publish in the patient education section of a healthcare website." The simplified PEMs were read by the authors to confirm accuracy. Readability was then reassessed. The mean RGL of the 27 simplified articles was determined and was compared with 6th grade and 8th grade reading levels, using one-sample t-tests. The mean RGL of the original articles was compared to that of the simplified articles using an independent samples *t*-test. All statistical analysis was carried out in Jamovi (The jamovi project (2024). jamovi (Version 2.5) [Computer Software]. Retrieved from https://www.jamovi.org).

3. Results

3.1. Readability of hand and wrist patient education materials

A total of 135 PEMs were identified from the three websites. The mean RGL for all 135 articles was 9.89 (range = 7.2–15) (Table 1; Fig. 1). The mean RGL was 9.73 (95 % CI, 9.48–9.98) for the ASSH website; 10.2 (95 % CI, 9.9–10.5) for the AAOS website; and 10.0 (95 % CI, 9.4–10.6) for the BSSH website. There was no significant difference between the RGLs of the three websites ($F = 0.854$; $p = 0.43$).

Examining the overall mean RGL for each article showed that no article was written at or below the recommended 6th grade reading level and only 4 articles (2.96 %) were written at or below the average reading level of 8th grade. The overall mean RGL of the 135 articles significantly exceeded the 6th grade level by 3.89 grade levels (95 % CI, 3.7–4.08; $P < 0.001$) and the 8th grade level by 1.89 grade levels (95 % CI, 1.7–2.08; $P < 0.001$). Table 2 and Fig. 2 show the breakdown of mean RGL by readability formula. Of the 135 PEMs, 33 (24.4 %) could not be evaluated via the Fry test and 43 (31.9 %) via the Raygor Estimate, due to the use of too many complex words. The mean Flesch-Kincaid Reading Ease (FRES) index was 60.5, which is just above the minimum of 60 to be considered plain English.

3.2. ChatGPT 4o as a tool to simplify hand and wrist patient education materials

The original 27 articles had a mean RGL of 10.4 (95 % CI 9.87–10.9). This was significantly greater than both the recommended level of 6th grade ($p < 0.001$) and the average level of 8th grade ($p < 0.001$). The overall mean RGL of the articles output from ChatGPT was 8.54 (Table 4), exceeding the requested 6th grade level by an average of 2.54 grade levels (95 % CI, 2.13–2.96; $P < 0.001$). The mean was higher than the 8th grade level by an average of 0.54 grade levels, however, this was not statistically significant (95 % CI, 0.126–0.956; $P = 0.013$). Tables 3 and 4 show the original and simplified articles respectively, divided by

Table 1
Patient education materials reading grade level mean by website.

Website	Number of articles	Mean Reading Grade (95 % CI)
ASSH	80 (59.3 %)	9.73 (9.48–9.98)
AAOS	32 (23.7 %)	10.2 (9.9–10.5)
BSSH	23 (17 %)	10 (9.4–10.6)
Overall	**135 (100 %)**	**9.89 (9.7–10.1)**

ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand.

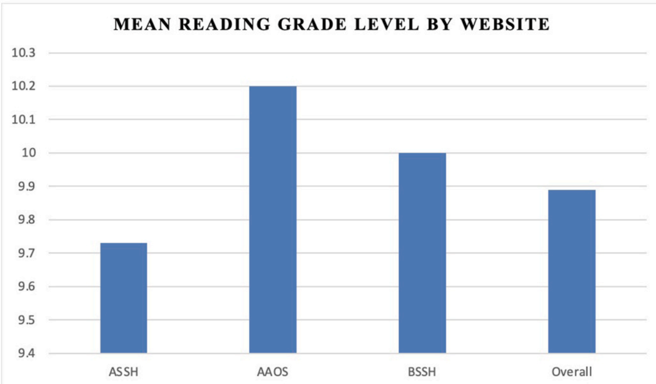

Fig. 1. Patient education materials reading grade level mean by website. ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the hand.

Table 2
Mean reading grade level by readability formula.

Test	ASSH (Range)	AAOS (Range)	BSSH (Range)
Flesch-Kincaid	7.99 (5.4–11.6)	8.41 (6.8–10.1)	8.56 (6.4–13.7)
Raygor Estimate[a]	9.14 (6–12)	10.1 (8–13)	9.9 (7–17)
Fry[a]	8.65 (6–12)	9.39 (7–12)	9.32 (7–17)
SMOG	10.7 (8.6–14.4)	11.1 (9.3–12.6)	11.1 (9.4–15.6)
Coleman-Liau	10.1 (6.8–13.5)	10.6 (8.5–12.7)	9.9 (8.3–13.6)
FORCAST	10.6 (9.1–11.9)	10.7 (9.8–11.9)	10.4 (9.4–11.6)
Gunning Fog	10.2 (7.2–15.1)	10.6 (8.2–12.8)	10.9 (8.4–16.5)
Mean	**9.73 (7.2–12.6)**	**10.2 (8.3–12)**	**10 (8.4–15)**

ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand.
[a] Some articles unable to be evaluated due to the use of too many complex words.

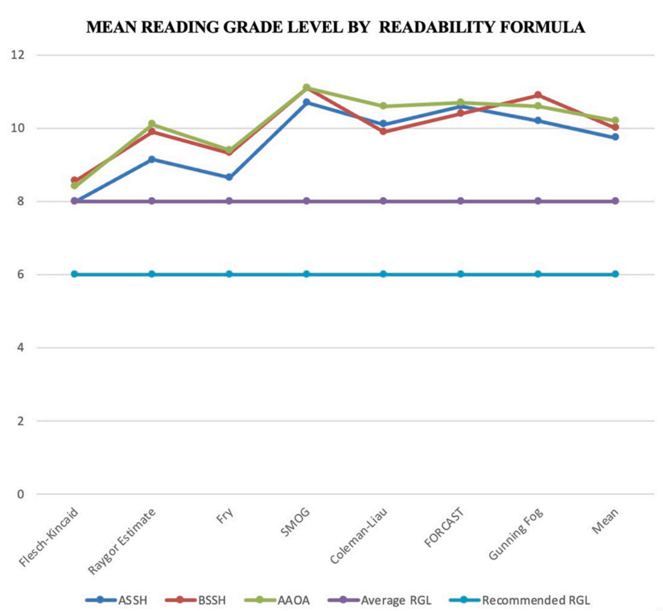

Fig. 2. Mean reading grade level by readability formula. ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand.

website. Table 5 shows the breakdown of the mean RGL of the simplified articles by readability formula. The means of the RGLs of the simplified articles were significantly lower than that of the original articles (95 %

Table 3
Mean reading grade of original articles by website.

Website	Number of articles	Mean Reading Grade (95 % CI)
ASSH	9 (33.3 %)	10.4 (9.48–11.3)
AAOS	9 (33.3 %)	10.4 (9.92–10.9)
BSSH	9 (33.3 %)	10.4 (8.86–11.8)
All Original	**27 (100 %)**	**10.4 (9.87–10.9)**

ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand.

Table 4
Mean reading grade of simplified articles by website.

Website	Number of articles	Mean Reading Grade (95 % CI)
ChatGPT ASSH	9 (33.3 %)	8.63 (7.84–9.43)
ChatGPT AAOS	9 (33.3 %)	8.68 (8.01–9.34)
ChatGPT BSSH	9 (33.3 %)	8.31 (7.32–9.30)
All ChatGPT	**27 (100 %)**	**8.54 (8.13–8.96)**

ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand.

Table 5
Mean reading grade level of simplified articles by readability formula.

Readability Formula	All ChatGPT Mean RGL (Range)
Flesch-Kincaid	6.86 (5.2–9.8)
Raygor Estimate[a]	7.58 (6–10)
Fry[a]	7.21 (6–10)
SMOG	9.63 (8.4–12.8)
Coleman-Liau	9.33 (7.3–12.1)
FORCAST	10 (9–11.4)
Gunning Fog	8.36 (6.2–12.5)
Mean	**8.54 (7–11.3)**

[a] Some articles unable to be evaluated due to the use of too many complex words.

CI, 1.2–2.49; $p < 0.001$). Fig. 3 shows the mean RGL of each original article compared to the corresponding simplified article. While the target of a 6th grade reading level was not reached, this study does show promising results on the ability of ChatGPT 4o to simplify PEMs.

4. Discussion

With increasing emphasis on shared decision-making and the internet being a primary source of health information for many patients, the accessibility of PEMs has become more important than ever.[1,4] Despite the well-documented association between poor health literacy and poor patient outcomes,[5,6] limited efforts have been made to simplify hand and wrist surgery PEMs to improve their readability. While the average reading grade level in the U.S. is around the 8th grade level, the NIH previously recommended a 6th-7th grade reading level for health materials, and the AMA advocated for materials written at or below a 6th grade level.[10,11] Despite these guidelines, hand and wrist surgery PEMs have consistently remained above the recommended levels.

Our study revealed that the mean RGL of the hand and wrist surgery PEMs from ASSH, AAOS, and BSSH was 9.89, which is significantly higher than both the recommended 6th grade level and the average U.S. reading level of 8th grade. A 2022 study by Zhang et al.[22] assessed the readability of ASSH PEMs and found a slightly lower average RGL of 9.8, which is comparable to our findings. The inclusion of PEMs from AAOS and BSSH in our study, along with the use of different readability formulae, may explain the minor differences between the two studies.

Our study also explored the potential of ChatGPT 4o to simplify PEMs in hand and wrist surgery. Recent studies, including those by Vallurupalli et al. and Oliva et al., have examined the use of ChatGPT 3.5 for improving PEM readability, with promising results.[20,23,24] Unlike the

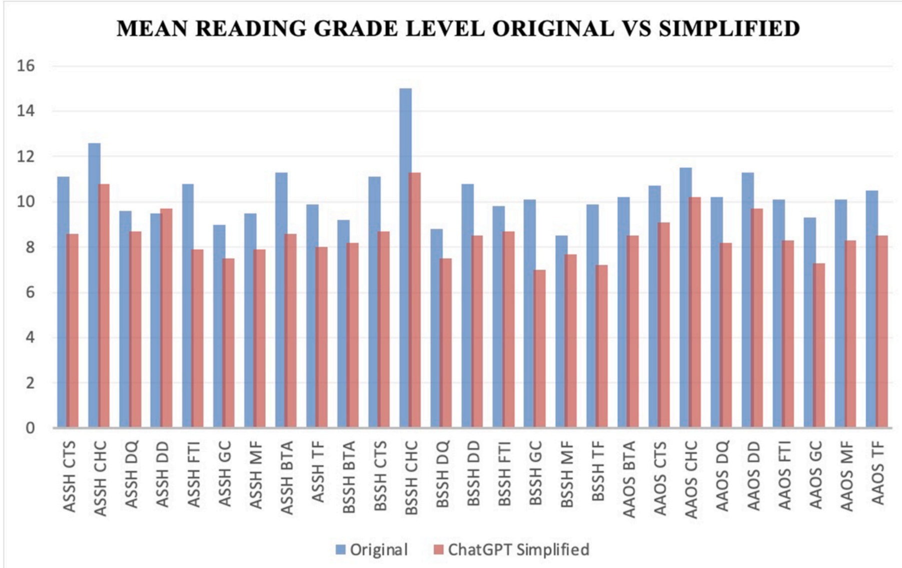

Fig. 3. Mean Reading Grade Level of Original vs Simplified Articles. ASSH: American Society for Surgery of the Hand; AAOS: American Academy of Orthopaedic Surgeons; BSSH: British Society for Surgery of the Hand CTS: Carpal Tunnel Syndrome; CHC: Congenital Hand Conditions; DQ: De Quervain's syndrome; DD: Dupuytren's Disease; FTI: Flexor Tendon Injury; GC: Ganglion Cyst; MF: Mallet Finger; BTA: Basal Thumb Arthritis; TF: Trigger Finger.

studies from Vallurupalli et al. which simplified excerpts from PEMs, our study assessed the entirety of each PEM, aiming to better reflect real-world applications. We observed a statistically significant reduction in RGL, from 10.4 to 8.54, which aligns with the recent results from Vallurupalli et al. and Oliva et al.

While our study did not achieve the target 6th grade reading level, the results indicate that ChatGPT 4o shows promise in simplifying medical information for patients. The ChatGPT-simplified PEMs consistently demonstrated better readability than the original materials, suggesting that ChatGPT could be a valuable tool for healthcare professionals involved in drafting or revising PEMs. However, it remains clear that further refinement is necessary to fully meet the established readability standards. Future research could consider requesting AI to simplify to a lower RGL, such as 5th grade, to assess whether the output material would be closer to the recommended 6th grade level. In addition, future research could consider assessing the efficacy of other large language models and generative AI such as Google's Gemini.

There are several limitations to this study that should be considered. Although ChatGPT demonstrated potential for simplifying text and reducing medical jargon, there is a risk of oversimplification, potentially leading to the omission of crucial medical details. This highlights the need for medical professionals to review the AI-generated text to ensure accuracy and that the pertinent information has not been removed. Additionally, ChatGPT 4o's knowledge is limited to information available up to 2023, which may result in outdated content. This emphasises the importance of oversight to ensure accurate and up-to-date information.

Whilst the eight chosen readability formulae are widely used in education and have been used in previous PEM readability studies, their application in healthcare literature has not been fully validated. These formulae assess readability based on factors like word length and sentence structure. Still, they may not account for the difficulty posed by medical terminology, which could be unfamiliar to patients despite being linguistically simple. Another potential limitation lies in the selection of PEMs. We analysed materials from only three major organisations – ASSH, AAOS, and BSSH – and limited our analysis to English-language articles. While these are prominent sources and likely to be recommended by surgeons, they may not represent all the PEMs that patients access online.

In conclusion, hand and wrist surgery PEMs produced by leading healthcare organisations continue to have readability scores that are above the recommended levels. Although ChatGPT 4o shows promise in simplifying PEMs and improving their accessibility, professional oversight is still essential to ensure accuracy and completeness. Moving forward, AI tools like ChatGPT could play a key role in bridging the gap between current readability levels and health literacy recommendations, but further research is needed to optimise these tools for widespread clinical use.

Guardian/patient's consent

Patient or guardian consent was not required for this study as it did not involve human participants or identifiable patient data.

Credit author statement

Ailbhe Quigley: Conceptualisation, Methodology, Investigation, Data curation, Resources, Formal analysis, Visualisation, Writing – original draft preparation.

Tiarnán Ó Doinn: Conceptualisation, Methodology, Resources, Formal analysis, Supervision, Writing – reviewing and editing.

Louise Kent: Investigation, Validation, Data curation, Writing – reviewing and editing.

James M. Broderick: Supervision, Writing – reviewing and editing, Project administration.

Ethical approval

Ethical approval was not required for this study as it involved analysis of publicly available, non-identifiable patient education materials and did not include human participants or animal subjects.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT in order to simplify the Patient Education Materials to examine ChatGPT's ability to improve readability. ChatGPT was also used during manuscript preparation to check spelling and grammar. The authors reviewed and edited the content and take full responsibility for the publication's content.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Appendix 1. Patient Education Materials Included in Analysis

American Academy of Orthopaedic Surgeons (AAOS)

1. Adult Forearm Fractures
2. Arthritis of the Hand
3. Arthritis of the Thumb
4. Arthritis of the Wrist
5. Boutonnière Deformity
6. Brachial Plexus Injuries
7. Carpal Tunnel Syndrome
8. Complex Regional Pain Syndrome (Reflex Sympathetic Dystrophy)
9. Congenital Hand Differences
10. De Quervain's Tenosynovitis
11. Distal Radius Fractures (Broken Wrist)
12. Dupuytren's Disease
13. Erb's Palsy (Brachial Plexus Birth Palsy)
14. Finger Fractures
15. Fingertip Injuries and Amputations
16. Flexor Tendon Injuries
17. Forearm Fractures in Children
18. Frostbite
19. Ganglion Cyst of the Wrist and Hand
20. Hand Fractures
21. Hook of the Hamate Fracture of the Wrist
22. Human Bites
23. Kienböck's Disease
24. Mallet Finger
25. Nerve Injuries in the Hand and Fingers
26. Scaphoid Fracture of the Wrist
27. Sprained Thumb
28. Thumb Fractures
29. Trigger Finger
30. Ulnar Tunnel Syndrome of the Wrist
31. Upper Extremity Limb Length Discrepancy
32. Wrist Sprains

American Society for Surgery of the Hand (ASSH)

1. Amputation: Prosthetic Hand And Fingers
2. Animal Bite
3. Arm Cast and Splint Care
4. Arm, Hand And Finger Replantation
5. Boutonniere Deformity
6. Boxer's Fracture
7. Brachial Plexus Injury
8. Broken Finger Tip
9. Broken Hand
10. Carpal Boss
11. Carpal Tunnel Syndrome
12. Cigarette Smoking and Hand Conditions
13. Cold Hands
14. Congenital Hand Differences
15. Congenital Trigger Thumb
16. Cortisone Shot
17. Cubital Tunnel Syndrome
18. Cyclist's Palsy
19. De Quervain's Tenosynovitis
20. Dressings After Hand Injury Or Surgery
21. Dupuytren's Contracture
22. Edema Treatment For The Arm And Hand
23. Extensor Tendon Injury
24. Fibromyalgia Hand Pain
25. Flexor Tendon Injury
26. Fractures in Children
27. Frostbite in Hands
28. Gamer's/Texter's Thumb
29. Ganglion Cyst
30. Golf injuries to the Hand, Wrist and Elbow
31. Gout in Hands
32. Hand Cramps (Focal Dystonia)
33. Hand Infection
34. Hand Surgery Anaesthesia
35. Hand Therapy
36. Hand Therapy for Hypersensitivity
37. Hand and Finger Exercises
38. Heat Treatment And Cold Treatment For Hands
39. How to Treat a Burn
40. Jammed Finger
41. Jersey Finger
42. Knuckle, Wrist & Finger Joint Replacement
43. MCP Joint arthritis
44. Mallet Finger
45. Nail Bed Injury
46. Nerve Damage and Repair
47. Nerve Injury
48. Numbness in Hands
49. Osteoarthritis
50. Pain Management: How to Get Pain Relief
51. Pain Medication: What are Opioids?
52. Paraffin Wax Bath
53. Pseudogout
54. Psoriatic Arthritis
55. Radial Tunnel Syndrome
56. Rheumatoid Arthritis
57. Rock Climbing Injuries
58. Scapholunate Torn Ligament
59. Scar Treatment
60. Skiing And Snowboarding Injury Prevention
61. Skin Cancer Of The Hand And Upper Extremity
62. Splinters and Other Foreign Bodies in the Hand
63. Sprained Thumb
64. Stiff Hands
65. Swan Neck Deformity
66. Swollen Fingers
67. Systemic Disease
68. Tendon Transfer Surgery
69. Thumb Arthritis
70. Trigger Finger
71. Vascular Disease
72. Warts on Hands
73. Wrist or Hand Tumor
74. Kienbock's Disease
75. Scaphoid Fracture
76. TFCC Tear
77. Ulnar Wrist Pain
78. Wrist Arthtitis
79. Wrist Fracture

80. Wrist Surgery: Arthroscopy

British Society for Surgery of the Hand (BSSH)

1. De Quervain's syndrome
2. Trigger finger/thumb
3. Ganglion cysts
4. Carpal tunnel syndrome
5. Cubital tunnel syndrome
6. Basal thumb arthritis
7. Terminal finger joint arthritis
8. Dupuytren's disease
9. Congenital Hand Conditions
10. Boutonniere injury
11. Extensor tendon injury
12. Flexor tendon injury
13. Mallet finger injury
14. Thumb extensor tendon (EPL) rupture
15. General information on hand fractures
16. Finger dislocations
17. Finger sprains
18. Skier's thumb
19. Volar plate injury
20. Nerve injury
21. Hand wounds
22. Nailbed injuries
23. Wrist sprains

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Abbreviations

AAOS: American Academy of Orthopaedic Surgeons
AAO-HNS: American Academy of Otolaryngology-Head and Neck Surgery
AHRQ: Agency for Healthcare Research and Quality
AI: Artificial Intelligence
ALA: American Laryngological Association
AMA: American Medical Association
ANOVA: Analysis of Variance
ASSH: American Society for Surgery of the Hand
BOA: British Orthopaedic Association
BSSH: British Society for Surgery of the Hand
BTA: Basal Thumb Arthritis
ChatGPT: Chat Generative Pre-trained Transformer
CHC: Congenital Hand Conditions
CTS: Carpal Tunnel Syndrome
DD: Dupuytren's Disease
DQ: De Quervain's Syndrome
FKGL:: Flesch-Kincaid Reading Grade Level
FRES: Flesch Reading Ease Score
FTI: Flexor Tendon Injury
GC: Ganglion Cyst
MF: Mallet Finger
NIH: National Institute of Health
PEM: Patient Education Material
RGL:: Reading Grade Level
SMOG: Simple Measure of Gobbledygook
TF: Trigger Finger